

Fall 2021

Problem 1 (Fa21). Find a particular solution y of $y' = y^2 - ty + 1$ and then the general solution.

Problem 2 (Fa21). Let

$$g(x) = \sin^2(\pi x) + \sin^2(\pi x) \cos^2(\pi x) + \sin^2(\pi x) \cos^4(\pi x) + \cdots + \sin^2(\pi x) \cos^{2k}(\pi x) + \cdots$$

Evaluate the limit

$$G(x) = \lim_{n \rightarrow \infty} g(n!x)$$

Is G Riemann integrable?

Problem 3 (Sp77, Su82, Fa18, Fa21). Suppose f is a differentiable real valued function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$ and $f(0) = 0$. Prove that $f(x) > 0$ for all positive x .

Problem 4 (Fa14, Fa15, Fa21). 1. Suppose $p(z) = c(z - a_1) \cdots (z - a_n)$ is a complex polynomial. If z has positive real part and all the zeros a_i have negative real part, show that $p'(z)/p(z)$ has positive real part.

2. Show that all the zeros of the derivative p' of a complex polynomial lie in the convex hull of the zeros of p .

Problem 5 (Fa77, Fa81, Fa21). Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by $u(x, y) = x^3 - 3xy^2$. Show that u is harmonic and find $v : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that the function $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by

$$f(x + iy) = u(x, y) + iv(x, y)$$

is analytic.

Problem 6 (Fa21). Solve the initial value problem

$$y' = Ay = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} y \quad y(0) = y_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Problem 7 (Fa21). Let the $n \times n$ complex matrix A have entries

$$A_{pq} = \exp(-2it_p t_q)$$

where t_p is real for $1 \leq p \leq n$. Suppose A has the *Singular Value decomposition* $A = U \Sigma V^*$ i.e., U and V are unitary and Σ is diagonal and non-negative. Find a Singular Value decomposition of the matrix B with entries $B_{pq} = \exp(i(t_p - t_q)^2)$

Problem 8 (Su85, Fa21). Let G be a group of order 120, let H be a subgroup of order 24, and assume that there is at least one left coset of H (other than H itself) which is equal to some right coset of H . Prove that H is a normal subgroup of G .

Problem 9 (Fa94, Fa21). Let p be an odd prime and $\mathbb{F}_p$ the field of p elements. How many elements of $\mathbb{F}_p$ have square roots in $\mathbb{F}_p$? How many have cubic roots in $\mathbb{F}_p$?

Problem 10 (Sp20, Fa21). Find the unique continuous extension g of $f(x) = x^x$ to the half-open interval $\mathbb{R}_+ = [0, \infty)$ and determine all its fixed points there. Describe the limit x_∞ of the sequence $x_{n+1} = g(x_n)$ as a function of $x_0 \in \mathbb{R}_+$.

Problem 11 (Sp20, Fa21). Let $p(t) = t^n - p_1 t^{n-1} - p_2 t^{n-2} - \dots - p_n$ where all the coefficients p_j are positive. Show that p has exactly one positive zero.

Problem 12 (Fa21). Suppose all the zeros x_j of the n th degree polynomial

$$p(x) = x^n - a_1 x^{n-1} + a_2 x^{n-2} - \dots$$

are real. Show that the largest zero has absolute value at most $\sqrt{a_1^2 - 2a_2}$ and find all polynomials where equality holds.

Problem 13 (Su80, Fa17, Fa21, Fa24). Let f be a meromorphic function on $\mathbb{C}$ which is analytic in a neighborhood of 0. Let its Maclaurin series be

$$\sum_{k=0}^{\infty} a_k z^k$$

with all $a_k \geq 0$. Suppose there is a pole of modulus $r > 0$ and no pole has modulus $< r$. Prove there is a pole at $z = r$.

Problem 14 (Sp78, Sp18, Fa21). Show that there is a complex analytic function defined on the set $U = \{z \in \mathbb{C} \mid |z| > 4\}$ whose derivative is

$$f(z) = \frac{z}{(z-1)(z-2)(z-3)}$$

Is there a complex analytic function on U whose derivative is

$$g(z) = \frac{z^3}{(z-1)(z-2)(z-3)} ?$$

Problem 15 (Sp83, Fa21, Sp20, Fa20). Let A be an invertible real $n \times n$ matrix. Show that there is a decomposition $A = QR$ in which Q is an $n \times n$ real orthogonal matrix and R is upper triangular with positive diagonal entries. Is this decomposition unique?

Problem 16 (Fa21). Suppose that A and B are positive definite matrices. Show that if $A \geq B$, meaning $A - B$ is positive semidefinite, then $A^{-1} \leq B^{-1}$.

Problem 17 (Fa21). Let $G_1 \leq G_2 \leq \dots$ be an infinite sequence of groups, and

$$G = \bigcup_i G_i$$

Show that there is a group structure on G such that $G_i \leq G$ for all i .

Problem 18 (Fa21). Let $\frac{1}{1-x-x^2} = \sum_{n=0}^{\infty} f_n x^n$ be a Taylor series expansion convergent near $x=0$.

1. Find a and b such that $f_{n+1} = af_n + bf_{n-1}$ for $n \geq 1$.
2. Show that $\gcd\{f_{n+1}, f_n\} = 1$.